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Grading table

Question:	1	2	3	4	5	Total
Points:	10	8	8	5	7	38
Score:						

Instructions

- Answer all questions on the exam paper in the space provided beneath the question. Unjustified answers will receive little or no credit. If you need more space, continue on the back of the page, CLEARLY INDICATING THAT YOUR WORK IS TO BE CONTINUED.
- The value of each question is indicated beside the statement of the question. The total value of all questions is 38 points.
- No texts, notes, or other aids are permitted. There are no calculators, cellphones, or any other electronic device permitted.
- Do not remove staples or any pages from the exam.
- If you have any questions, feel free to ask me.
- Good Luck!



1. (10 points) Use mathematical induction to prove that for any integer $n \geq 1$

$$1 + 3 + 5 + 7 + \cdots + (4n + 1) = (2n + 1)^2.$$



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2. (8 points) Use induction to prove that 5 divides $9^{2n} - 6^n$ for any integer $n \geq 1$.



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3. (a) (5 points) Use the formula $1 + 3 + 5 + 7 + \cdots + (4n + 1) = (2n + 1)^2$ of question 1 to evaluate the sum

$$\sum_{k=2}^{12} (2k + 3).$$

Simplify your result as much as possible.

- (b) (3 points) Write the sum $-1 + 2 + 5 + 8 + 11 + 14 + \cdots + 41$ in sigma notation starting at the integer 1. Do not evaluate.



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4. (5 points) Evaluate the expression

$$i^{44} \frac{1+3i}{1-3i} + 2 \cdot i^{17} + (1-i)^2.$$

The solution should be in Cartesian form and simplified as much as possible.



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5. (7 points) Find all the real numbers t for which the complex number

$$z = 6 \frac{it - 1}{2 - it}$$

has imaginary part equal to 1.



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